SOFTWARE ENGINEERING PROJECT PRESENTATION

Section - 541

MEMBERS

Sai Lyeng Kham	6631013
Peerapong Chaichumpoo	6631014
Nyunt Tin	6632004

BinGO

An application for waste management system

Problem Statements

Fixed schedules create inefficient waste collection.

Improper disposal contaminates and fills landfills.

Poor routing drives up operational expenses.

Lack of up-to-date data limits optimization.

STAKEHOLDERS

- citizens
- waste management workers
- recycling center operators
- IT staff
- application developers/designers
- third-party inspectors

Project Goals

Citizen reporting via mobile app for full bins and waste hotspots.

Data-driven optimization of pickup schedules and collection routes

In-app education on proper waste disposal practices

Reduction of operational costs through efficient collection trips

Enhanced environmental sustainability and increased recycling

SCOPE

CITIZENS

- Registration/Login with emails
- Image and message report submission
- Displayed proper waste disposal tips
- Provided information and support from Recycling centers
- Submitted reports tracking
- Notifications for announcements, waste pickup schedules and community clean-up events

SCOPE

WORKERS

- Log in with their organization accounts
- Received filled bins and trash accumulation reports
- Optimized collection routes to improve efficiency
- Status of waste collection task updates
- Notifications and alerts on high-priority waste areas
- Communicate with citizens regarding waste collection concerns
- Generated reports on collection efficiency and operational challenges
- Access to a dashboard summarizing citywide waste reports

SCOPE

RECYCLING CENTER

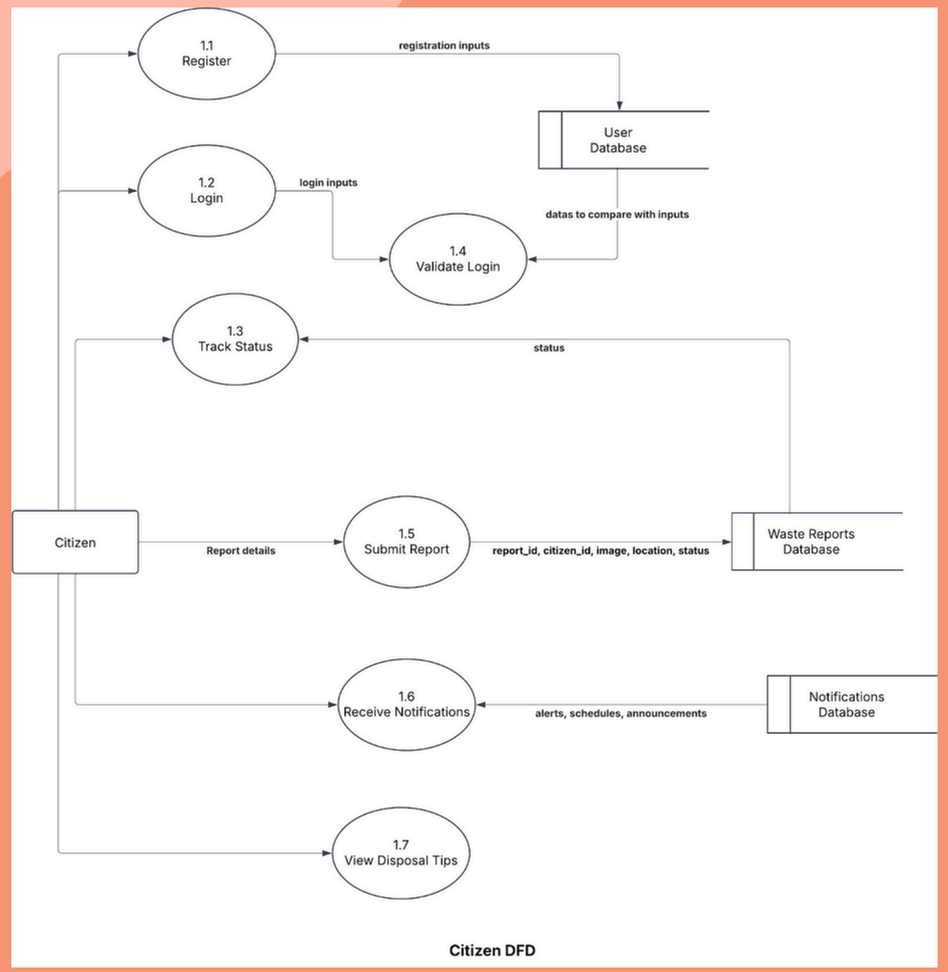
- Log in with their organization accounts
- Can receive data on recyclable waste reports from citizens.
- Can list recycling guidelines for public awareness.
- Can notify citizens about recyclable waste collection schedules.
- Can coordinate with waste management workers for efficient waste sorting.

SCOPE

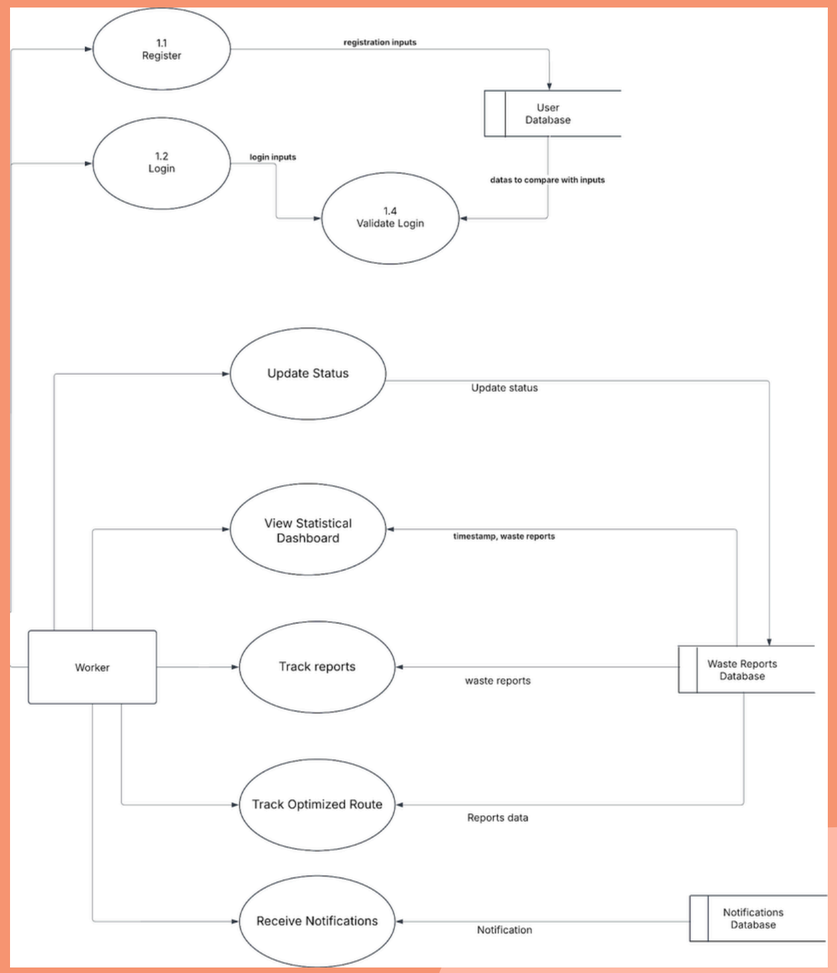
ADMIN

- Can log in with an administrator account.
- Can oversee the overall application system and user management.
- Can manage and approve citizen-submitted waste reports.
- Can update the waste management schedule and optimize system workflows.
- Can analyze system data for insights on waste trends.
- Can generate and export reports for city officials and stakeholders.
- Can implement new features and updates based on feedback.

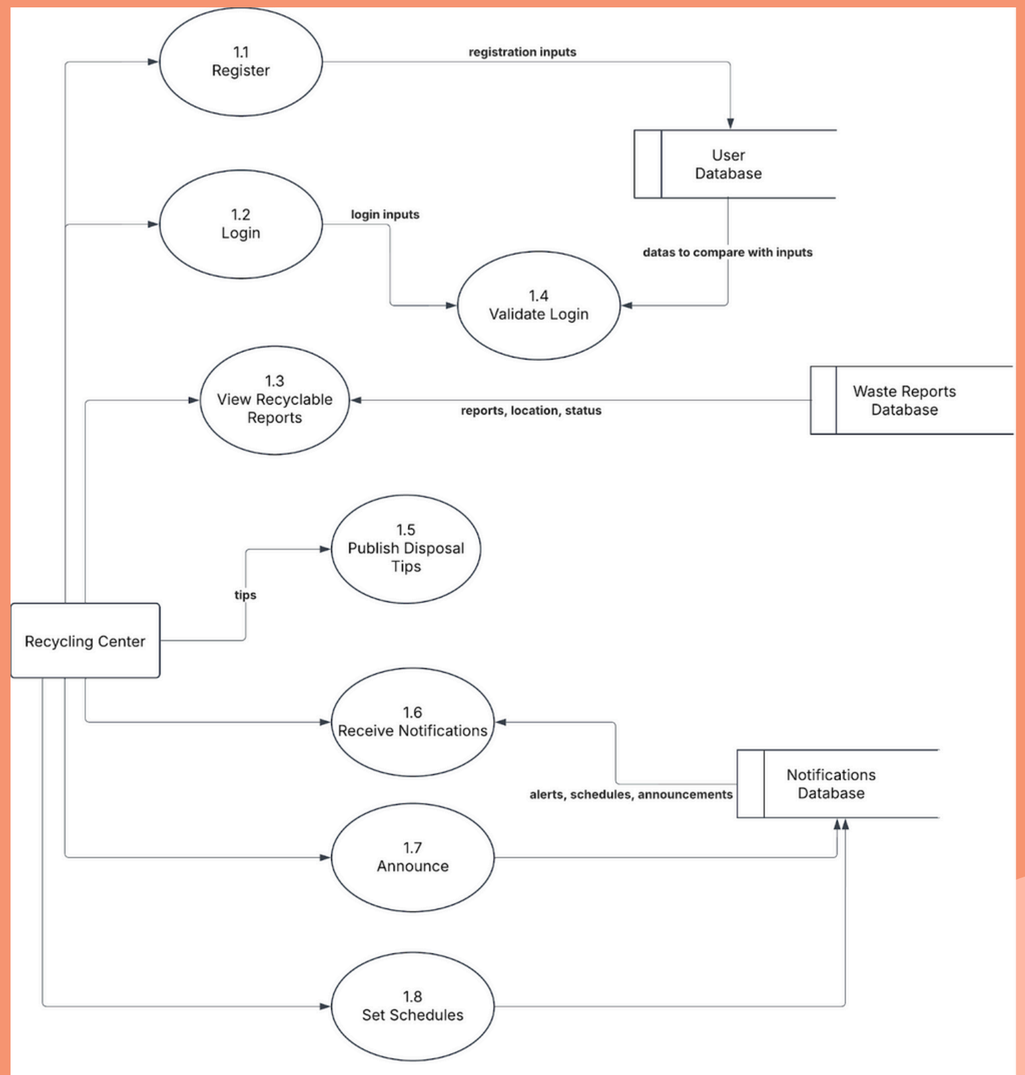
DFD Diagram



Citizen DFD

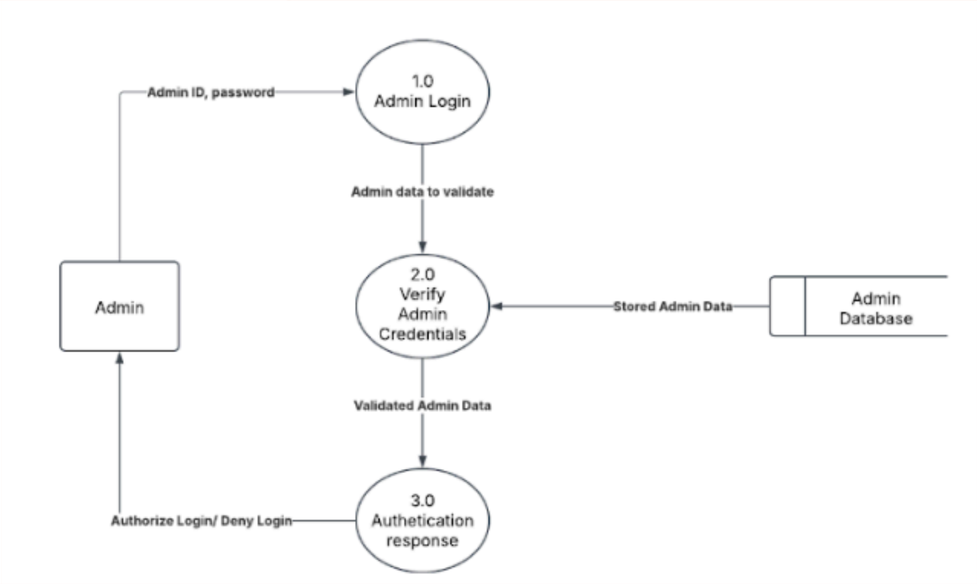


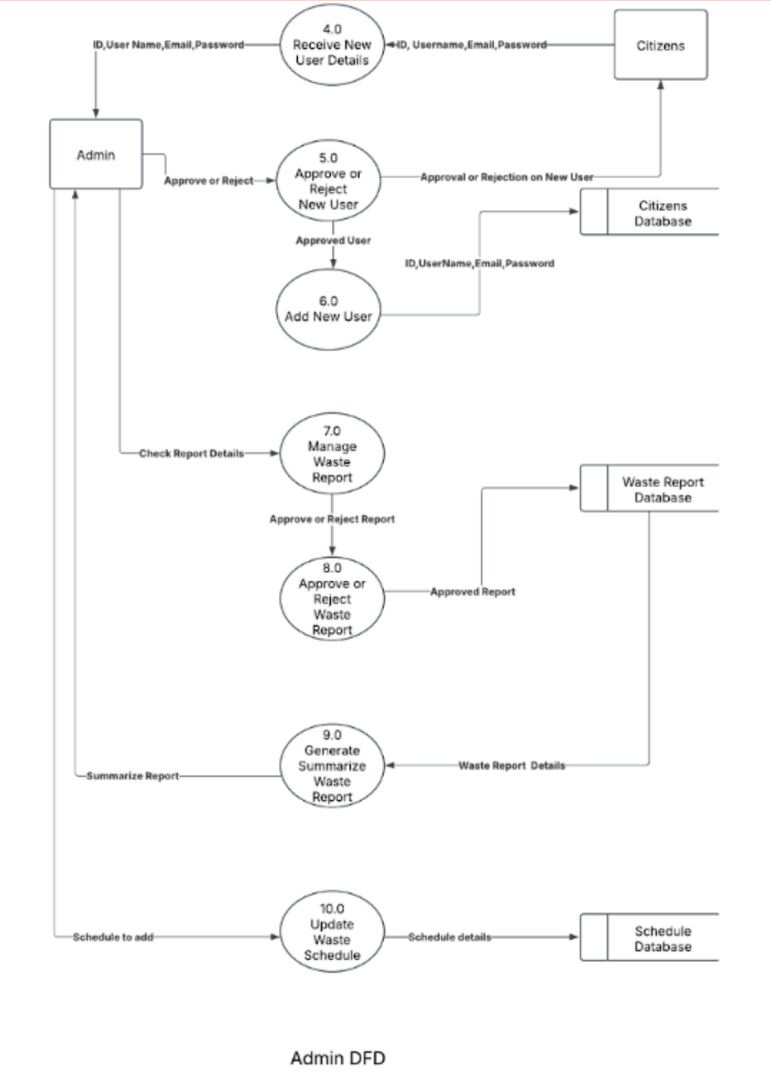
DFD Diagram



Recycling Center DFD

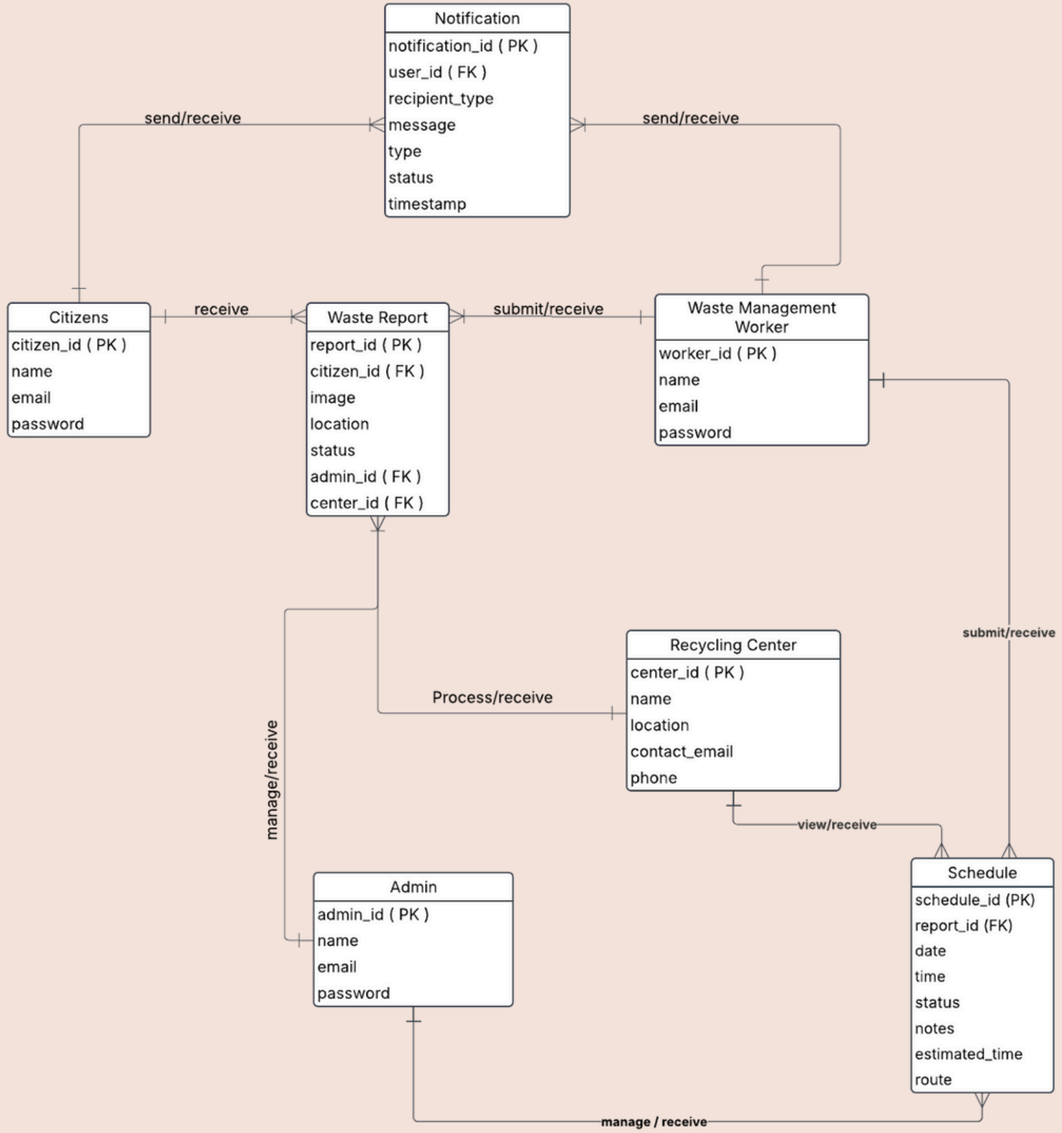
DFD Diagram





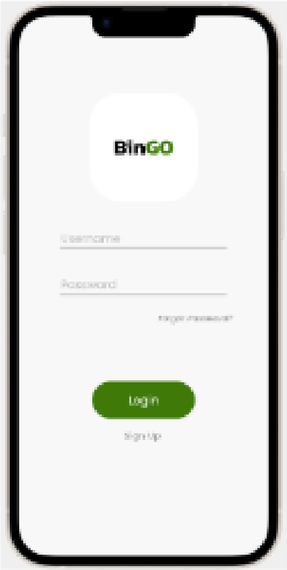
Admin DFD

ER Diagram

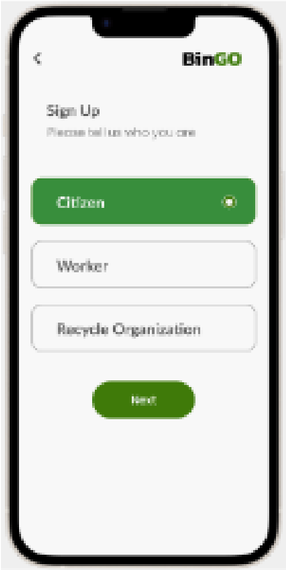

Notification
notification_id (PK)
user_id (FK)
recipient_type
message
type
status
timestamp
send/receive
send/receive
Citizens
citizen_id (PK)
name
email
password
receive
Waste Report
report_id (PK)
citizen_id (FK)
image
location
status
admin_id (FK)
center_id (FK)
submit/receive
Waste Management Worker
worker_id (PK)
name
email
password
submit/receive
Recycling Center
center_id (PK)
name
location
contact_email
phone
Process/receive
manage/receive
view/receive
Admin
admin_id (PK)
name
email
password
Schedule
schedule_id (PK)
report_id (FK)
date
time
status
notes
estimated_time
route
manage / receive

Project Prototype

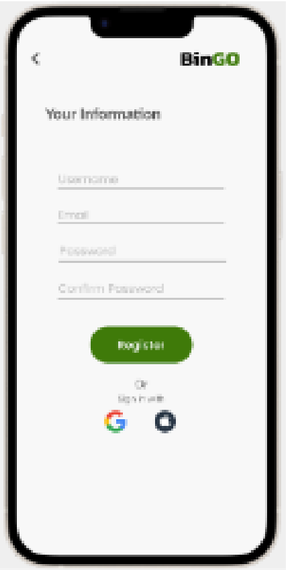
Project Prototype (User Interface)



Login Screen

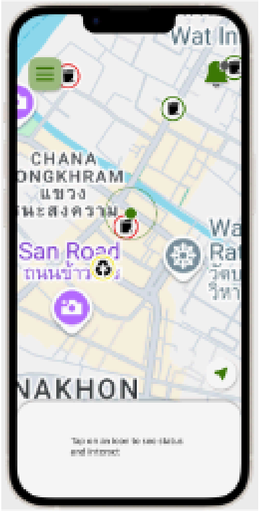


Register 1

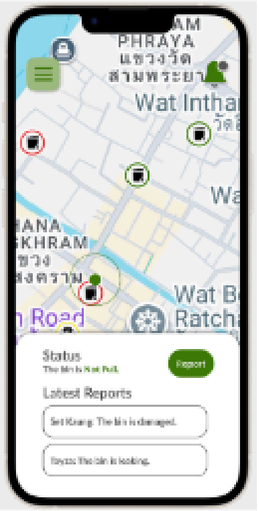


Register 2

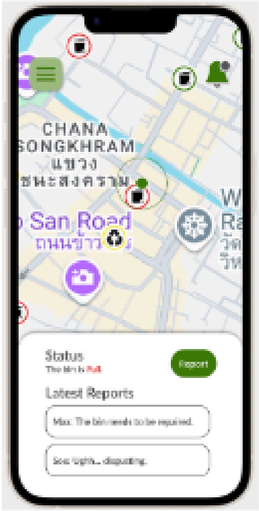
Citizen Interface



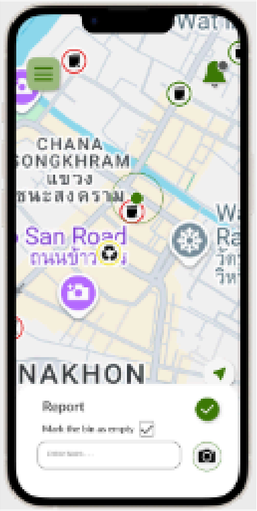
Home Page



Status(Not Full)

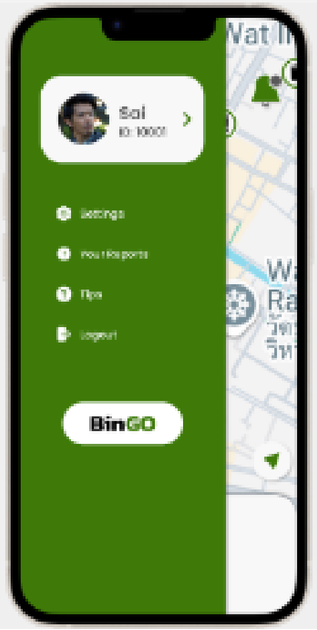


Status(Full)

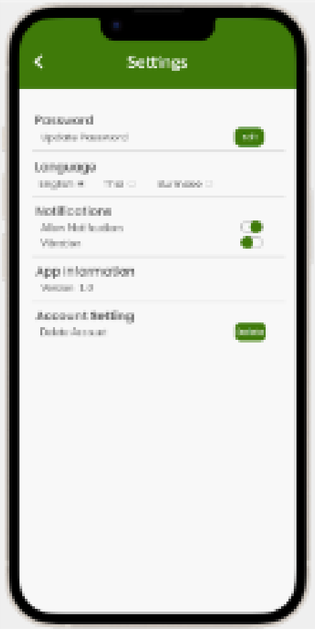


Report

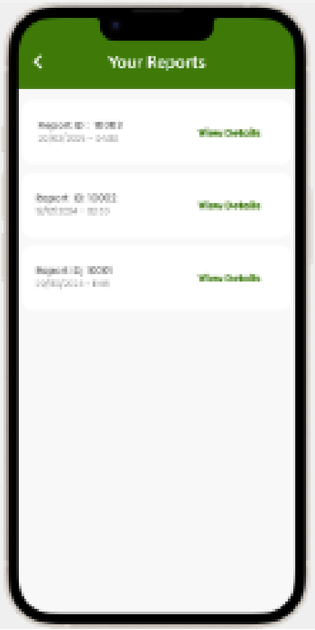
Project Prototype



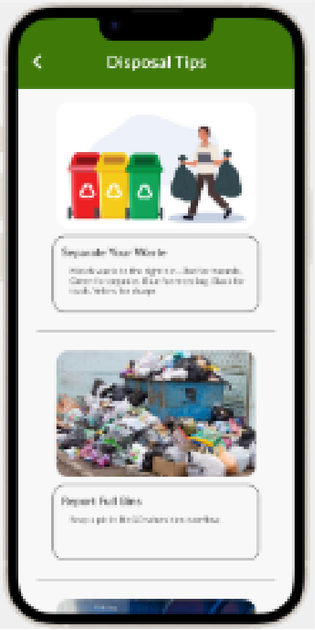
Menu



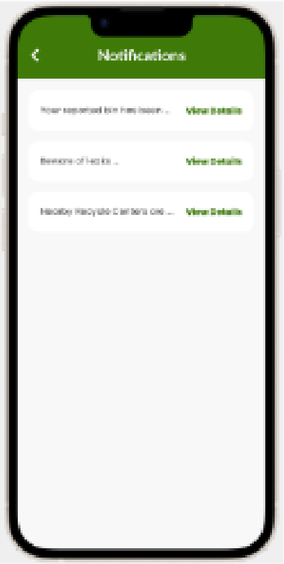
Settings



Your Reports



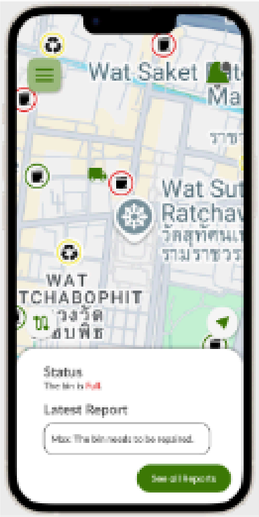
Disposal Tips



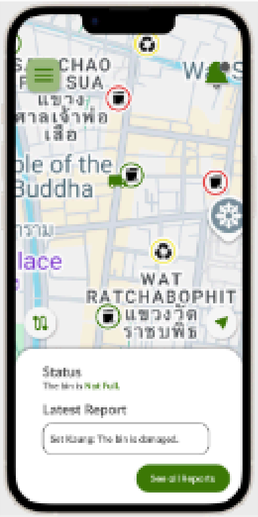
Notifications

Project Prototype

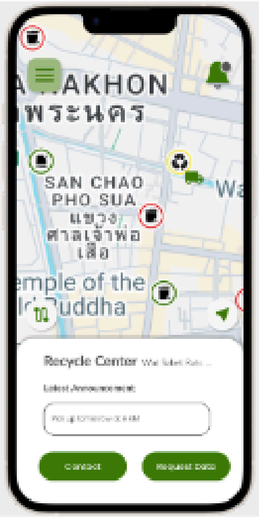
Worker Interface



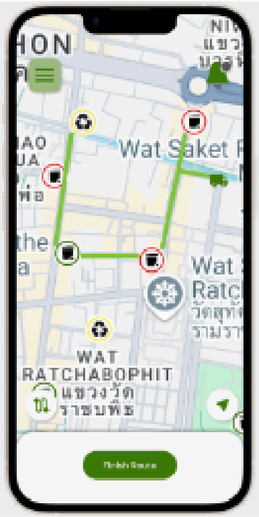
Status (Full)



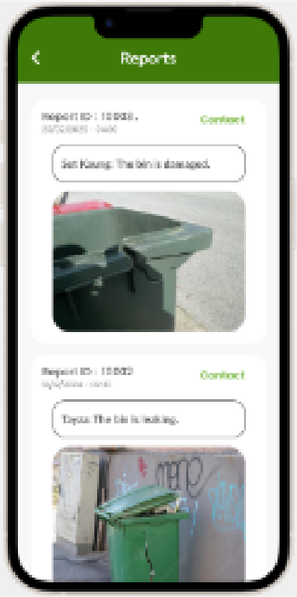
Status(Not Full)



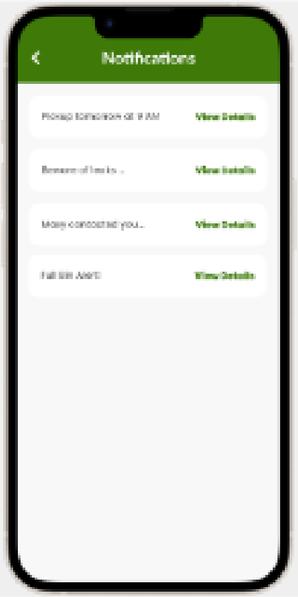
Status(Recycle Center)



Organized Routing



Reports



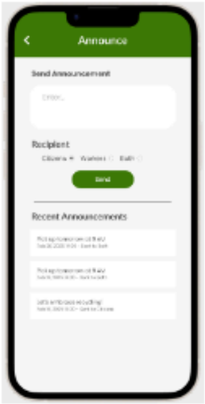
Notifications

Dashboard

Project Prototype

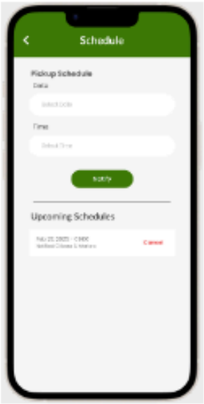
Recycle Center Interface

Home Page

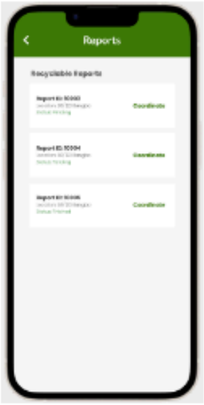


Announce

Tips



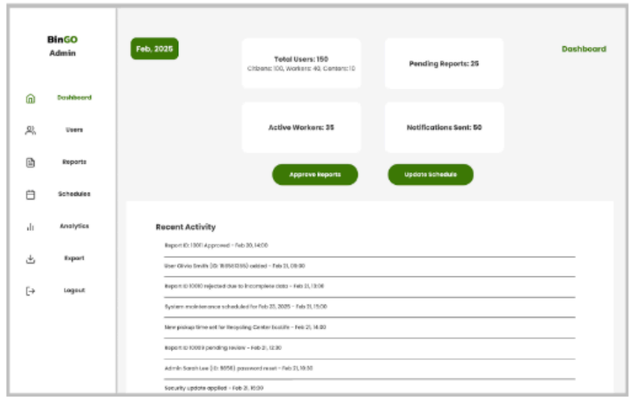
Schedule



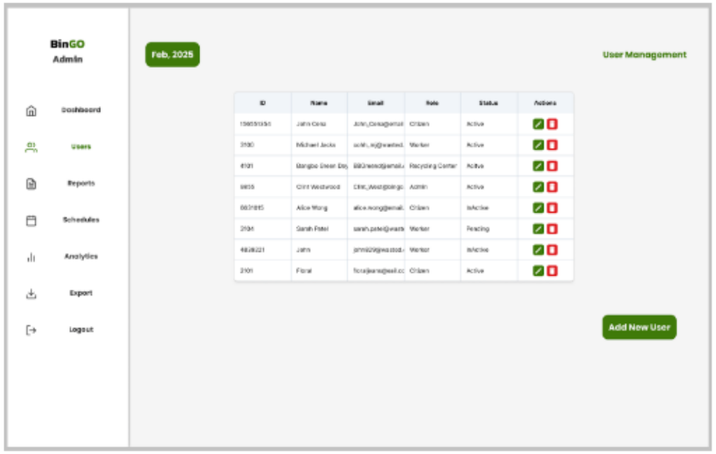
Reports

Project Prototype

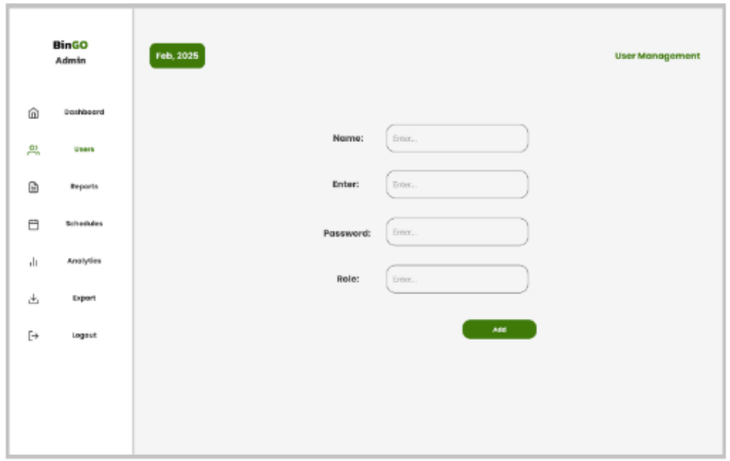
Admin Interface



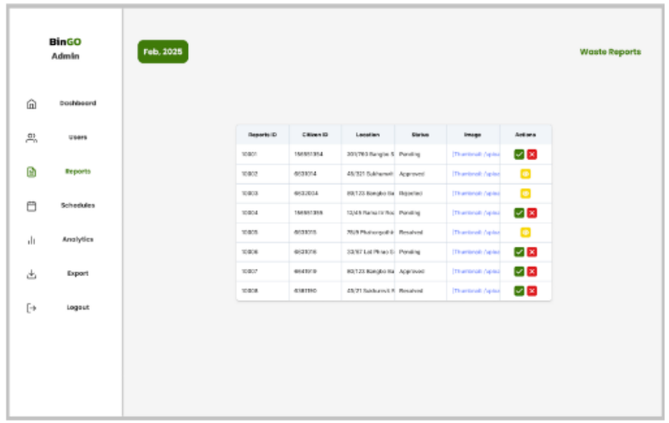
Dashboard



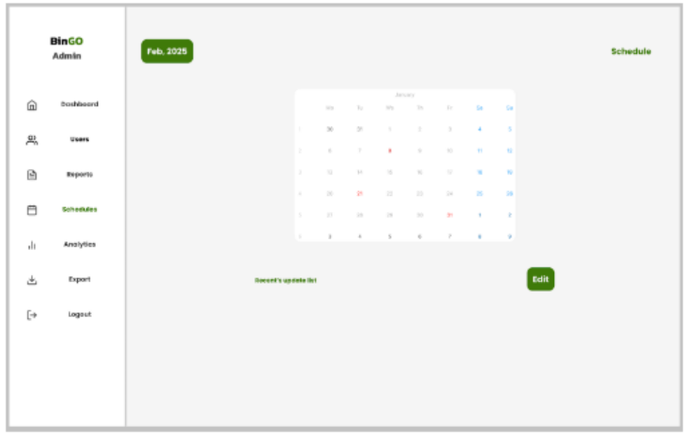
Users



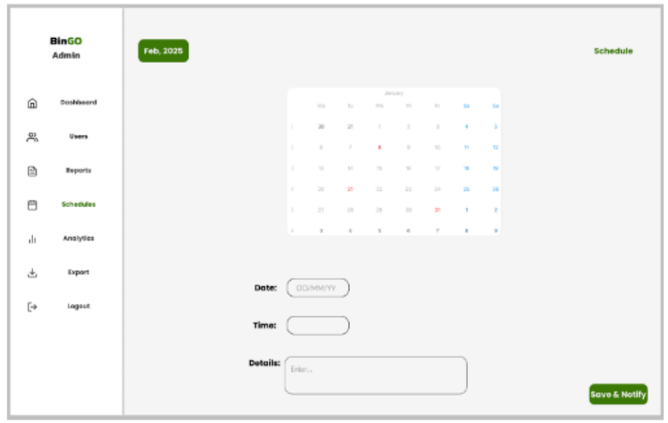
Add new user



Reports

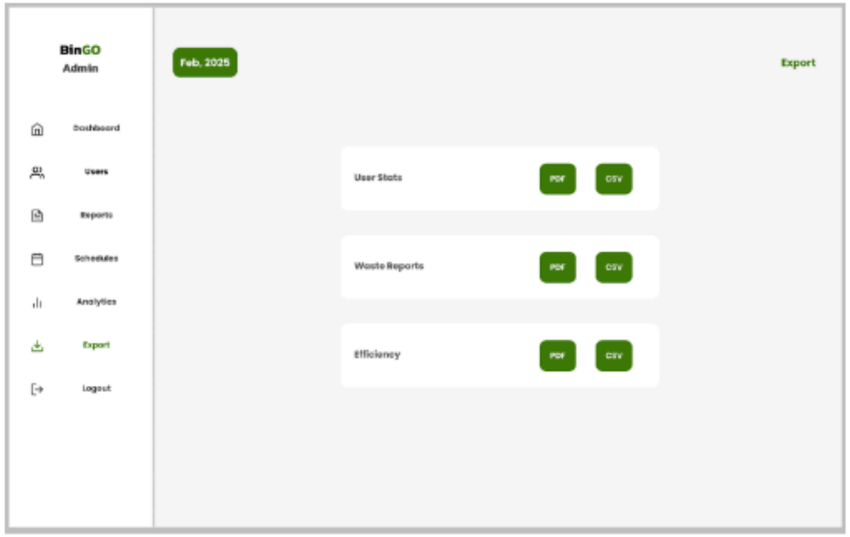


Schedules



Edit Schedule

Project Prototype



Export

Analytics

Project Development Steps (Scrum-Based Agile Process)

1. Project Planning
2. Requirement Analysis
3. Prototype Development
4. Sprint Planning
5. Sprint Review
6. Sprint Retrospective
7. Iteration
8. Project Finalization

Development Tools

IDE: IntelliJ IDEA – Powerful IDE for JavaScript & Node.js, ideal for full-stack development.
Version Control:
Git – Distributed version control system.
GitLab – Repository hosting & CI/CD integration.
Frontend Development: React Native – Cross-platform mobile app framework.
Backend Development: Django – Python-based framework for backend & API development.
Database Management: Firebase – Real-time database for prototyping & user data.
Cloud Services: MongoDB Atlas (M10) – Cloud database services.
Authentication: Firebase Authentication – Secure user authentication.
Project Management & Collaboration: Microsoft Teams – Team communication platform.
UI/UX Design: Figma – Collaborative design & prototyping tool.

Project Timeline

Project Timeline

Phase	Duration	Start (Relative)	Milestones/Sprints
Initiation/Planning	1 month	Project Start	Requirements Gathering, System Design, Project Plan Finalized
Development	6 months	+1 month	
* Sprint 1	2 weeks		User Authentication & Account Management - Registration/Login
* Sprint 2	2 weeks		User Authentication & Account Management - Profile Management
* Sprint 3	2 weeks		Waste Report Submission - Form Design & Data Capture
* Sprint 4	2 weeks		Waste Report Submission - Image Upload & Geolocation
* Sprint 5	2 weeks		Waste Report Tracking - Basic Status Display & Filtering
* Sprint 6	2 weeks		Waste Report Tracking - Map Integration for Report Visualization
* Sprint 7	2 weeks		Waste Worker Management - Basic User Interface & Role Setup
* Sprint 8	2 weeks		Waste Worker Management - Assignment of Reports to Workers
* Sprint 9	2 weeks		Basic Reporting & Analytics - Data Aggregation & Simple Charts
* Sprint 10	2 weeks		Basic Reporting & Analytics - Report Generation & Export
* Sprint 11	2 weeks		User Feedback & Support - In-App Feedback Form
* Sprint 12	2 weeks		User Feedback & Support - FAQ/Help Section
* Sprint 13	2 weeks		System Optimization - Performance Improvements & Bug Fixes
* Sprint 14	2 weeks		System Optimization - Security Enhancements & Code Refactoring
Testing/Quality Assurance	1 month	+7 months	System Testing, Integration Testing, Bug Fixing
Deployment/Launch	1 month	+8 months	Deployment to Production, User Acceptance Testing, Go-Live

Cost Estimation

Total Estimated Cost

Item	Cost	Note
Employee Cost	**1,395,000 THB**	**3 Developers**
*Junior Developer	(included)	35,000 THB/Month
*Mid-level Developer	(included)	50,000 THB/Month
*Senior Developer	(included)	70,000 THB/Month
Cloud Service	**76,500**	**MongoDB Atlas (M10)**
Development Tools	**36,000**	**GitHub Team Plan & Jira Software**
* GitHub Team Plan	(Included)	1,500 THB/month
* Jira Software	(Included)	2,500 THB/month
Miscellaneous	**50,000**	**Licenses, Testing Devices, etc.**

Total Estimated Cost **1,557,500 THB**

Cost Analysis

Cost Analysis
Estimated LOC: ~10,000 lines (10 KLOC).

Effort Calculation
KLoc (Thousands of Lines of Code) = 10
Formula: $E = 2.4 \times (KLoc)^{1.05}$
Calculation: $E = 2.4 \times (10)^{1.05} = 26.49$ person-months

Development Time Calculation
Formula: $D = 2.5 \times (E)^{0.38}$
Calculation: $D = 2.5 \times (26.49)^{0.38} = 8.33$ months (rounded to 9 months for practical purposes).

Number of People Required
Formula: $N = E / D$
Calculation: $N = 26.49 / 8.33 \approx 3$ persons

Benefits Analysis

- **Enhanced Efficiency –** Optimizes waste collection by improving resource allocation and response times.
- **Streamlined Problem Reporting –** Allows citizens to easily report overflowing bins and illegal dumping for faster resolution.
- **Data-Driven Decision-Making –** Provides real insights for better planning and resource allocation.
- **Reduced Unreported Cases –** Ensures fewer waste issues go unaddressed with a simple reporting mechanism.
- **Stronger Community Engagement –** Encourages direct communication between citizens and waste authorities.
- **Improved Transparency –** Enables users to track report progress, ensuring accountability.
- **Better Sanitation Standards –** Leads to overall improvements in public cleanliness and hygiene.
- **Valuable Waste Analytics –** Collects data on waste trends and hotspots for long-term planning.
- **Encouraging Social Responsibility –** Promotes citizen involvement in sustainable waste management.

THANK YOU